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Hiring Manager
Alstom Inc.
5 Sheridan Cl
Sydney, NSW 2214

Dear Hiring Manager,

I am writing to express my strong interest in the graduate engineering role at Alstom. As a fourth year mechanical engineering student at the University of Sydney, I am very excited about the prospects of contributing to the transition to fully renewable transport.

In my role as part of the aerodynamics department in the Formula-SAE (FSAE) team at the university, I have honed my CAD skills in SOLIDWORKS as well as numerical validation skills in ANSYS Fluent. My proficiency in these tools has allowed me to contribute significantly to our understanding of the aerodynamics package on our race cars, leading to a measurable lap time improvements.

Having been actively involved in the team's manufacturing processes, I am well versed in Australia's drawing standards, and thanks to my ability to work collaboratively with a diverse team across multiple areas of the car, manage complex engineering projects and achieving the design goals set, I have been nominated as one of the lead aerodynamicists. This experience has allowed me to understand how to design parts easily manufactured and maintained in low-volume applications such as FSAE while achieving the design goals in mind.

Alstom's commitment to sustainability and cutting-edge technology really draws me to this role, and I am extremely interested in optimizing certain aspects of the project such as the thermal cooling solution, exterior design and such to maximize the performance of the vehicle. I would like to use my skills in computational design and validation to address any design challenges.

I am confident that my hands-on experience in numerical computation, manufacturing, proficiency in industry-standard software, and proven ability to work collaboratively makes me a valuable candidate for this opportunity. I am eager to bring my skills and passion for engineering design to Alstom and would welcome an opportunity to discuss how I can contribute to the team's success.

Thank you for considering my application, I look forward to the possibility to contributing to the exciting innovations at Alstom.

Sincerely,
Winston Zhang

EXPERIENCE

SYDNEY MOTORSPORTS

Lead Aerodynamicist

Sydney, Australia

2022.07 - Current

DOVETAIL ELECTRIC AVIATION

Computational Fluid Dynamics Intern

Sydney, Australia

2023.09 - Current

TRANSFERABLE SKILLS

Problem Solving

Rapid response to critical failures during composite manufacturing of aerodynamic surfaces to ensure the part is made and minimal waste is produced. Resolving design challenges that are presented while minimizing weight and cost.

Communication

Technical details from other departments within the team need to be communicated clearly in order to ensure the design produced is realistic and will work together.

Time Management

Ensuring that personal design deadlines are met for the Formula SAE team as well as Dovetail Electric Aviation while studying full time.

TECHNICAL SKILLS

Composite Manufacturing

Part of the manufacturing of a aerodynamics package for Sydney Motorsports through both wet layup and prepreg. The molds were made from high density foam shaped by a variety of equipment such as CNC hot wire cutting, CNC routing and hand sculpting.

Computer Aided Design

Modified parts of the aerodynamic package for Sydney Motorsports as well as creating the baseline simulation geometry for Dovetail Electric Aviation. All CAD projects were made to be fully parameterized to allow for simple iterations of the design.

Numerical Simulation

Wrote a semi-automated workflow for testing aerodynamic component designs over a range of angles for sensitivity studies. Created a simple CFD model of a aircraft engine assembly including mesh analysis study including simulating a radiator in order to study the cooling effects. Conducted a thermal study in both fluent and Steady State Thermal of how much the heat output of a human and two computers would heat up a room.

Programming

Capable of writing analysis code in MATLAB and Python. Created a blade element study in MATLAB to study the performance of various propellers' performance under different flight conditions. Wrote custom FEM code and compared it with the commercial Steady State Thermal package.

PROJECTS

Semi-Automated CFD Workflow

Worked with and improved the robustness of the previous semi-automated CFD workflow in Sydney Motorsports through experimenting with the mesh settings. This allows more designs to be iterated through the workflow.

Aerodynamic Package for FSAE Car

Involved planning the manufacturing, mounting and validation of the second in house aerodynamics package for Sydney Motorsports. Multiple relatively simple and low cost methods are planned such as utilizing foam core and carbon honeycomb sandwich structures to increase the ease of manufacturing and strength of the components produced. Mounting solutions are improved to make track-side assembly and maintenance significantly faster.

Custom Cloud Storage Server

Created a custom server from e-waste and second hand parts for self-hosted cloud storage solutions similar to OneDrive with a RAIDZ5 configuration. The system also acts as a media server and NAS for additional storage, and can be access remotely through VPN.

UNIX OS

Dual booted various distros of UNIX such as Ubuntu and Arch alongside Windows on my PC out of interest for light-weight OS. This allowed me to be comfortable with TUI commands as well as familiarity with VIM.

Topology Optimization

Optimized the design of a rib and spar system for the front wing of the FSAE car with different materials to find a balance between performance, cost, manufacturing and time constraints.

EDUCATION

THE UNIVERSITY OF SYDNEY

Bachelor of Mechanical Engineering Honours (Fluid Mechanics) - Credit WAM

2021 - Current